

INTRO TO DYNAMIC PROGRAMMING

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A TYPICAL PROBLEM

- Many problems seen in economics have a common structure:
 - We observe the current **state** X_t at time t .
 - We choose an **action** A_t at time t .
 - We get a **reward** R_t at time t .
 - The state progresses to X_{t+1} at time $t + 1$.
- If the largest possible t is $T < \infty$, then we have a **finite horizon** problem. Otherwise we have an **infinite horizon** problem.

EXAMPLE

- You saw a simple lifecycle model with known income path (*perfect foresight*).
- It was easy to solve: just calculate optimal consumption and savings in every period.
Possible to do it by hand.
- What if income earned by households is uncertain?
- What if households can invest in assets with uncertain returns?
- We need to find optimal solution for all possible paths of income and returns.

EXAMPLE

- Consider a problem of a firm that produces a good. The firm wants to maximize the expected present discounted value of profits:

$$\mathbb{E} \sum_{t=0}^{\infty} \left(\frac{1}{1+r} \right)^t \pi_t$$

- X_t is the current state of the firm. It can be the current level of capital, the current level of inventory, prices set by competitors...
- A_t is the action taken by the firm. It can be the level of production, the level of future inventory, the price of the good...
- R_t is the reward. Here it is the profit of the firm, π_t .
- X_{t+1} is the state of the firm in the next period. It can depend on the current state X_t and the action A_t taken.

EXAMPLE

- This is potentially an **extremely** complicated problem.
- For example: the state X_t can include the demand for the good – and it could be random.
- Find actions for all possible future states...
- We will learn tools that can help us solve such problems.

PLAN

- Today we will study an example: **McCall's job search model (1970)**.
- Exposition based on Stachurski and Sargent (2023).
- Next time: more general theory of dynamic programming.

MCCALL'S JOB SEARCH MODEL

TWO-PERIOD PROBLEM

- An unemployed agent receives a job offer at wage W_t .
- She can either **accept** the offer or **reject** it.
- If she **accepts**, she gets this wage permanently.
- If she **rejects**, she gets unemployment benefit c .
- Wage offers are independent and identically distributed (i.i.d.) and nonnegative, with distribution ϕ :
 - $W \subset \mathbb{R}_+$ is a finite set of possible wages.
 - $\phi : W \rightarrow [0, 1]$ is a probability mass function, $\phi(w)$ is the probability of getting a wage w .
- The agent is risk-neutral and impatient: the utility of getting y tomorrow is βy , with $\beta \in (0, 1)$.

TWO-PERIOD PROBLEM

- The agent lives for two periods ($t = \{1, 2\}$) and starts unemployed.
- The question: is it better to accept a received offer or wait until tomorrow hoping for better offer?
- What is the lowest wage that the agent should accept?
- We will start analyzing the problem by looking at the second period, $t = 2$: **backward induction**.

PERIOD $T=2$

- Suppose the agent is unemployed at $t = 2$.
- She gets a wage offer W_2 which she can either **accept** or **reject** the offer.
- **Accept**: get income $W_2 \rightarrow$ get utility W_2 .
- **Reject**: get income $c \rightarrow$ get utility c .
- Since this is the last period of her life, she will accept if and only if

$$W_2 \geq c.$$

PERIOD T=1

- The agent gets a wage offer W_1 .
- She can either (a) accept and get W_1 forever, or (b) reject and get c in period $t = 1$ and then get $\max \{W_2, c\}$ in period $t = 2$.
- The utility of (a) is $W_1 + \beta W_1$. We call it the **stopping value**.
- The utility of (b) is $h_1 := c + \beta \mathbb{E} \max \{W_2, c\}$. We call it the **continuation value**.

$$h_1 = c + \beta \sum_{w' \in W} v_2(w') \phi(w'), \quad v_2(w') := \max \{w', c\}$$

- The agent will accept if and only if the **stopping value** is greater than the **continuation value**:

$$W_1 + \beta W_1 \geq h_1.$$

VALUE FUNCTION

- The key object in dynamic programming is the **value function**.
- It is a **function** that maps the state to the **maximum** expected present discounted value of future rewards.
- In our example, the **state** at each time is the received wage offer w .
- $v_2(w)$ is the value function at time $t = 2$ and wage w : the largest possible reward that the agent can get if she starts unemployed at $t = 2$ and gets a wage offer w . We have

$$v_2(w) = \max \{w, c\}.$$

- The time 1 value function is

$$v_1(w) := \max \left\{ (1 + \beta) w, c + \beta \sum_{w' \in W} v_2(w') \phi(w') \right\}.$$

TWO-PERIOD EXAMPLE

- This particular problem is easy to solve.
- The agent accepts if and only if the stopping value exceeds the continuation value:

$$(1 + \beta) w \geq h_1 \iff w \geq \frac{h_1}{1 + \beta}.$$

- So the value function is

$$v_1(w) = \begin{cases} (1 + \beta) w & \text{if } w \geq \frac{h_1}{1 + \beta} \\ h_1 & \text{otherwise.} \end{cases}$$

- In context of this example, we call $w^* := \frac{h_1}{1 + \beta}$ the **reservation wage**.
- We see that since h_1 is increasing in c , the reservation wage is higher when the unemployment benefit c is higher.

THREE-PERIOD EXAMPLE

- Extend the model by one period, $t = 0$.
- The value function at $t = 0$ is

$$v_0(w) := \max \left\{ w + \beta w + \beta^2 w, c + \beta \sum_{w' \in W} v_1(w') \phi(w') \right\}.$$

where the formula for v_1 is from the previous slide.

- **Key insight:** at $t = 0$ it is like a two-period problem.
- **All** information about the future is summarized in v_1 , the value function at $t = 1$.
- This is the standard approach: convert a complicated dynamic optimization problem into a sequence of two-period problems.

BELLMAN EQUATION

- Recall we had

$$v_2(w) = \max \{w, c\}$$

$$v_1(w) = \max \left\{ w + \beta w, c + \beta \sum_{w' \in W} v_2(w') \phi(w') \right\}$$

$$v_0(w) = \max \left\{ w + \beta w + \beta^2 w, c + \beta \sum_{w' \in W} v_1(w') \phi(w') \right\}.$$

- The recursive relationships between the value functions are called the **Bellman equations**.
- Warning:** these equations are (in general) **functional equations**. We need to find functions, not numbers. Here it is easy: we have a finite set of possible wages – treat functions as vectors.

INFINITE HORIZON

- The three-period problem is also easy to solve.
- In fact, we can use the same approach (backward induction) as before for any finite horizon problem.
- What if the horizon is **infinite**? We no longer have the **terminal** period.
- Dynamic programming makes this problem **tractable**.

INFINITE HORIZON

- The objective function is

$$\mathbb{E} \sum_{t=0}^{\infty} \beta^t R_t,$$

where R_t is either c (unemployed) or W_t (employed).

- Let $\beta \in (0, 1)$ be the discount factor, as before we let $c > 0$.
- The wage process satisfies $(W_t)_{t \geq 0} \stackrel{iid}{\sim} \phi$ where $\phi \in \mathcal{D}(W)$ and $W \subset \mathbb{R}_+$ with $|W| < \infty$.
- For any finite or countable set F , $\mathcal{D}(F)$ is the set of distributions on F .

INFINITE HORIZON

- What is **stopping value**?
- If the worker accepts wage w she gets

$$w + \beta w + \beta^2 w + \beta^3 w + \dots = \frac{w}{1 - \beta}.$$

- What is the **continuation value**?
- If the worker rejects wage w she gets

$$c + \beta \sum_{w' \in W} v(w') \phi(w')$$

where the value function v is the same in all periods – the problem is **stationary** (always infinite remaining future).

INFINITE HORIZON

- Bellman equation is:

$$v(w) = \max \left\{ \frac{w}{1 - \beta}, c + \beta \sum_{w' \in W} v(w') \phi(w') \right\}.$$

- **Principle of optimality, Bellman (1960):** *An optimal policy has the property that whatever the initial state and the initial decisions it must constitute an optimal policy with regards to the state resulting from the first decision.*
- This is not that trivial, we will return to it (and prove it!) later.
- Interpretation: value function satisfies the Bellman equation.

CHALLENGE

- Bellman equation is:

$$v(w) = \max \left\{ \frac{w}{1 - \beta}, c + \beta \sum_{w' \in W} v(w') \phi(w') \right\}.$$

- Once we have $v(w)$ we can characterize the optimal choice of the agent.
- Q: how to find $v(w)$? It is a function!
- Q: is there a solution?
- Q: is the solution unique?
- Q: what are the properties of the solution?
- A: we will learn how to answer these questions.

APPROACH I

- This particular problem is relatively easy.
- Recall how we defined the **continuation value**:

$$h^* := c + \beta \sum_{w' \in W} \max \left\{ \frac{w'}{1 - \beta}, h^* \right\} \phi(w')$$

- We can write the value function as

$$v(w) = \max \left\{ \frac{w}{1 - \beta}, h^* \right\}.$$

- **Key:** h^* is a scalar, not a function. When would this break down?
- Find h^* directly, by solving the equation numerically.
- Then use h^* to get the value function.

APPROACH I

Algorithm Solving for v directly

```
1: procedure MCCALL
2:    $k \leftarrow 1, \epsilon \leftarrow \tau + 1, h_k \leftarrow c$ 
3:   while  $\epsilon > \tau$  do
4:      $h_{k+1} \leftarrow c + \beta \sum_{w' \in W} \max \left\{ \frac{w'}{1-\beta}, h_k \right\} \phi(w')$ 
5:      $\epsilon \leftarrow |h_{k+1} - h_k|, k \leftarrow k + 1$ 
6:   end while
7:   for  $w \in W$  do
8:      $v(w) \leftarrow \max \left\{ \frac{w}{1-\beta}, h_k \right\}$ 
9:   end for
10: end procedure
```

APPROACH II

- We know how to solve a finite horizon problem – backward induction.
- Maybe we can get an approximate solution to the infinite horizon problem by considering a finite horizon problem with a very large number of periods?
- We will later **prove** that it actually works.

BELLMAN EQUATION

- We want to solve the Bellman equation:

$$v(w) = \max \left\{ \frac{w}{1 - \beta}, c + \beta \sum_{w' \in W} v(w') \phi(w') \right\}.$$

- Introduce a **Bellman operator** T defined at $v \in \mathbb{R}^W$ as

$$(Tv)(w) := \max \left\{ \frac{w}{1 - \beta}, c + \beta \sum_{w' \in W} v(w') \phi(w') \right\}.$$

- Note: here we treat v as a vector, not a function. We can do it, because W is finite.
- Let $V := \mathbb{R}_+^W$ and let $\|\cdot\|_\infty$ be the supremum norm.

BELLMAN EQUATION

- There exists a unique fixed point v^* of T in V such that $v^* = Tv^*$.
- The fixed point of the Bellman operator solves the Bellman equation.
- We can obtain v^* by iterating the Bellman operator:

$$v_{k+1} = Tv_k, \quad k = 0, 1, \dots$$

- We can start with **any** $v_0 \in V$ and the sequence converges to v^* .
- **Mathematical details:** we will cover the theory behind this result in a separate lecture.

APPROACH II

Algorithm Value function iteration

```
1: procedure VFI
2:    $k \leftarrow 1, \epsilon \leftarrow \tau + 1, v_k \leftarrow v_{init}$ 
3:   while  $\epsilon > \tau$  do
4:     for  $w \in W$  do
5:        $v_{k+1}(w) \leftarrow (Tv_k)(w)$ 
6:     end for
7:      $\epsilon \leftarrow \|v_{k+1} - v_k\|_\infty, k \leftarrow k + 1$ 
8:   end while
9: end procedure
```

OPTIMAL CHOICES

- Once we have v^* we can characterize the optimal choice of the agent.
- We can calculate the continuation value h^* :

$$h^* := c + \beta \sum_{w' \in W} v^*(w') \phi(w').$$

- Reject the offer if $w / (1 - \beta) < h^*$, accept otherwise.
- Denote rejection given wage W_t as $A_t = 0$ and acceptance as $A_t = 1$.

OPTIMAL CHOICES

- Let $A_t = \sigma_t(W_t)$ be the optimal choice of the agent at time t given wage offer W_t .
- We call σ_t the (time t) **policy function**.
- In this particular case the policy function is

$$\sigma_t(w) = \begin{cases} 1 & \text{if } \frac{w}{1-\beta} \geq h^* \\ 0 & \text{otherwise.} \end{cases}$$

where $\sigma_t(w) = 1$ means accept and $\sigma_t(w) = 0$ means reject.

- The policy function here depends only on the current state (wage).
- We call such a policy function (depending on the current state only) **Markov policy**.

OPTIMAL CHOICES

- A policy is an "instruction manual" for the agent: what to do in each state.
- Let Σ denote the set of all policies (functions from wages to actions $\{0, 1\}$).
- For an agent following $\sigma \in \Sigma$, if the current wage offer is w , the agent will respond with $\sigma(w) \in \{0, 1\}$.
- For each $v \in V$, a **v -greedy policy** is a $\sigma \in \Sigma$ satisfying

$$\sigma(w) = \mathbb{1} \left\{ \frac{w}{1 - \beta} \geq c + \beta \sum_{w' \in W} v(w') \phi(w') \right\} \quad \text{for all } w \in W$$

where $\mathbb{1} \{\text{condition}\} = 1$ if condition is true and 0 otherwise.

- The recommendation is: **adopt a v^* -greedy policy** (notice the superscript!).
- This is a restatement of **Bellman's principle of optimality**.

LOOKING FORWARD

- Here we have a **finite** state space – we can treat v as a vector.
- We only used a fraction of the power of dynamic programming...
- **Next lecture**: mathematical foundations
 - What are the conditions under which the Bellman operator is a **contraction**? Is there an easy way to check it?
 - Does the **principle of optimality** hold in **general**?
- **Later**: moving beyond finite state and action spaces
 - Continuous state spaces
 - Function approximation methods